



Ocean

Software development framework

Release notes

Version 2024.1

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1. Overview

Ocean for Petrel is an open, proprietary application development framework for SLB Petrel E&P software platform. Ocean for Petrel provides a powerful software development environment through the extensive API footprint, through the .NET based architecture and the associated .NET tools.

The Petrel application is implemented on the Ocean framework and is bundled with Ocean and Ocean for Petrel. Ocean for Petrel enables access to the key Petrel domain data types, as well as user interface and interaction features. It also enables extensions of Petrel, allowing addition of new domain data types with renderers, new windows, and new algorithmic and interactive processes. Ocean for Petrel offers the opportunity to extend, interoperate with, and re-use many of the features of Petrel in custom workflows. This expedites delivery of technological innovation and significantly reduces the time-to-market.

2. A Request to our Clients

Your feedback on our API development is the primary key to our success, and thus we want to hear from you! Please feel free to send your comments to Ocean Portfolio.

Non-SLB Ocean users please pass your feedback via your contact person. The link above, as well as the links throughout this document, is for SLB internal use only. You can get more information about Ocean at <http://www.ocean.slb.com>. If you encounter defects or have suggested enhancements, you may also add them to mySLM Elite bug management system (or through your account representative). For myslm-elite.slb.com (internal SLB bug reports), use "Petrel" as the product line, "Infrastructure" or appropriate component name and the version.

If you have any issues, using new features, we are glad to assist. Please send a request to: <https://www.software.slb.com/support>.

3. Ocean for Petrel Distribution

Ocean for Petrel installation package is available for download through the SIS Software Download Center website and includes the following:

- Petrel 2024 (64 bit) installer and Ocean for Petrel API binaries
- Ocean for Petrel 2024 software development kit (SDK) documentation integrated with Visual Studio and in stand-alone Ocean for Petrel Help files (.chm); including:
 - Ocean for Petrel Wizards (Visual Studio 2019/2022 Plug-in)

- Sample Code and Reference Applications showing patterns for how to use the API

4. Product Licenses

Ocean for Petrel

Ocean is a licensed feature of the Petrel application. A Developer License is required to create new Ocean plug-ins for Petrel.

If you run Petrel 2024 with an Ocean for Petrel Developer License, all licensable features of the Petrel software will be available. The license may be used for testing the interoperability of developed Ocean modules with Petrel. The enabled Petrel features may not be used for any other purpose, such as offer or perform commercial services and perform commercial or competitive analysis of SLB's software. Furthermore, Petrel will configure itself to go into Developer's mode. This implies that the Petrel application shell and all hard copies will be watermarked. Any projects saved whilst in Developer's mode will be tagged. Such projects will continue to be watermarked even if you load these into a fully featured version of Petrel at a later point.

GeoToolkit.NET Development License Agreement

Vendor Run-Time Software	Unlimited GeoToolkit.NET run-time software for all SLB's SIS software including internal/external and university use; including the following libraries: WellLog.NET, Seismic.NET, Carnac.NET, Chart.NET, CGM output, WellSchematic.NET, Contour.NET and CarnacGIS.NET.
Vendor Development Software	Unlimited GeoToolkit.NET SDK licenses and maintenance fee for all End Users of SLB Software in relation to Ocean/Petrel development; including: Seismic.NET, Carnac.NET, Chart.NET, CGM output, WellSchematic.NET, Contour.NET and CarnacGIS.NET.

Open Inventor™ Development License Agreement

Vendor Run-Time Software	Unlimited OpenInventor.NET, MeshViz.NET, VolumeVizLDM.NET, Hardcopy.NET run-time software for a customer of SLB who is using the Ocean SDK covered by a current license agreement, or an internal SLB employee who is using the Ocean SDK for commercial and/or non-commercial use.
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Vendor Development
Software

Unlimited OpenInventor.NET, MeshViz.NET, VolumeVizLDM.NET, Hardcopy.NET SDK licenses and maintenance fee for all End Users of Ocean SDK in relation to Petrel or other SIS software development.

Microsoft .Net Compiler Platform ("Roslyn") Open Source License Notice

- Ocean for Petrel SDK Visual Studio extension wizard is developed using Roslyn under Apache 2.0 license, copy of this license is available is License.rtf file in installation folder.
- Roslyn source code is downloadable from GitHub ([//github.com/dotnet/roslyn](https://github.com/dotnet/roslyn)).
- Roslyn compiler platform is consumed as-is without support or maintenance.

5. Ocean for Petrel Documentation

The Ocean for Petrel help files (.chm) are generated from the source code at build time and is the ultimate source of API documentation. Please refer to this for the definitive list of API features.

6. Enhanced Security Practices: Important Information for Developers

In line with Microsoft's security guidelines available at <https://aka.ms/binaryformatter>, we strongly encourage developers to migrate away from BinaryFormatter, its associated types and unsafe alternatives to increase reliability and future-proof their work.

Implementations relying on these methods should start writing data with a preferred, safer alternative.

To persist your custom domain objects and your state information, our recommendation is to use the structured archive data source and the managed persistence APIs, as described in the documentation for "Custom Domain Objects" in the chapter "Managed persistence".

7. Operating Environment

- Ocean for Petrel 2024 (file version 24.1.0.x) has been validated on Windows 10 and Windows 11.
- Ocean for Petrel 2024 is built with:
 - Ocean Core version 1.5.19
 - Ocean Coordinate Services version 1.5.8
 - Ocean Geometry 1.5.2
- The following 3rd party software is needed for developing Ocean for Petrel modules:
 - Windows 10 64bit operating system or Windows 11 64bit operating system
 - Visual Studio Professional or Enterprise,
 - Visual Studio 2019 minimum version: Professional or Enterprise: 16.5
 - Visual Studio 2022 minimum version: Professional: 17.8 Enterprise 17.8
 - .NET Framework 4.8 Developer/Targeting Pack
 - Open Inventor™ and GeoToolkit.NET graphics are integral to the Ocean for Petrel API and the following versions are bundled with Ocean for Petrel 2024. See above for the terms of use.
 - Open Inventor 9.9.34.0
 - GeoToolkit.NET 20.0.8104.3
 - Infragistics 19.1 (assembly version Winforms 19.1.20191.238, WPF 19.1.20191.244). Unless a plug-in needs specific Infragistics control, the purchase of an Infragistics license is not required.
 - The Ocean for Petrel binaries were built with Visual Studio 2022. They target the .NET Framework 4.8.
 - Petrel 2024.1 requires minimum .NET Framework 4.8 at runtime.
 - Plug-ins consuming any of C++ 3rd party libraries shipped with Petrel will need to check against library version and the corresponding Visual C++ toolset requirement to ensure compatibility. In some cases, 3rd party library shipped with Petrel is run out of process..
 - The Ocean Plugin Installer wizard requires Wix toolset 3.11 or higher to be installed.

Petrel 2024.1 installs in parallel with earlier versions, so installers need to be updated for 2024.1 with the environment variable Ocean2024Home.

8. Petrel 2024.1 Features

Please refer to Petrel 2024.1 documentation for information regarding the features shipped with Petrel 2024.1 installation package.

9. Ocean API Stability

We are committed to compile-time compatibility in the Ocean API for a minimum of two full yearly releases. This enables Ocean Applications to easily migrate to subsequent commercial versions of Ocean for Petrel. API's that will be replaced and phased out in the future are first *deprecated* for at least one full yearly release (included in the overall two years minimum stability). Deprecated APIs will be marked [Obsolete] in the SDK documentation and your compiler will give warnings if you continue to use them; but we will continue to support and test them until they are removed.

There is an exception to this commitment, however: In some cases, we expect changes in the underlying API implementation that may require changes in the API. In these cases, we have tagged the API with the keyword [SubjectToChange] in the SDK documentation.

10. Plug-in Binary Compatibility

The Ocean for Petrel API assemblies are semantically versioned, according to the Major.Minor.Patch convention, defined as follows:

- MAJOR: Incompatible API changes
- MINOR: New API functionality that is backwards compatible
- PATCH: Bug fix that is backwards compatible

Therefore, a plug-in compiled for a lower minor version will be binary compatible with a higher version, but not the reverse.

Semantic versioning means that the version of an Ocean assembly may be lower than the Petrel version that deploys it. However, we ensure that Petrel loads plug-ins for as long as the API is backwards compatible. As a result, your plug-in may load and run in a higher major version of Petrel than it was originally built for. However, when a Major release of Petrel is announced, we would nonetheless advise the following:

- Check the latest version of Ocean. If there is a new Major version, you will need to produce an updated version of the plug-in
- Download the latest major versions of Petrel and the Ocean SDK and compile your plug-in. In addition to confirming (or otherwise) plug-in compatibility, this will inform you of any newly Obsolete API's

- Even if the plug-in builds without changes, we recommend precautionary testing

If the plug-in builds and loads without changes and you are satisfied with the outcome of your testing, your clients can install the same version of the plug-in with the new version of Petrel. However, if you produce a custom installer, you will need to build a new one.

Note that this binary compatibility statement applies only to the managed APIs in the Slb.Ocean namespaces as well as select 2nd and 3rd party managed APIs – OpenInventor (InventorNet.dll), GeoToolkit.NET, and the Microsoft Enterprise Library. If your plug-in depends on 3rd party libraries, you should check the Libraries table in the Quality assistant for conflicts with the versions deployed with Petrel 2024.1.

Note also that we may be unable to avoid changes to API behavior resulting from changes to the underlying behavior of Petrel (e.g., essential bug -fixes). Binary compatibility directly affects your clients' ability to deploy and maintain your plug-in so be aware of the version of Ocean for Petrel you compile against for your commercial release.

11. What's new in Ocean for Petrel 2024

Visual Studio 2022 Compatibility

Ocean for Petrel 2024.1 introduces support for Visual Studio 2022 and keeps support for Visual Studio 2019. However, Visual Studio 2017 is no longer supported.

12. Migrating from Visual Studio 2019 to Visual Studio 2022

When migrating from Visual Studio 2019 to Visual Studio 2022, the workflow for the NUnit Test Adapters has changed, as there is no NUnit Test Adapter Extension available for Visual Studio 2022.

As such, if you were using the NUnit Test Adapter Extension installed into Visual Studio 2019 to run tests, you may need to add the NUnit Test Adapter package from NuGet to your test projects after migrating to Visual Studio 2022.

To add NUnitTestAdapter to your test project, right-click your test project and select "Manage NuGetPackages..."

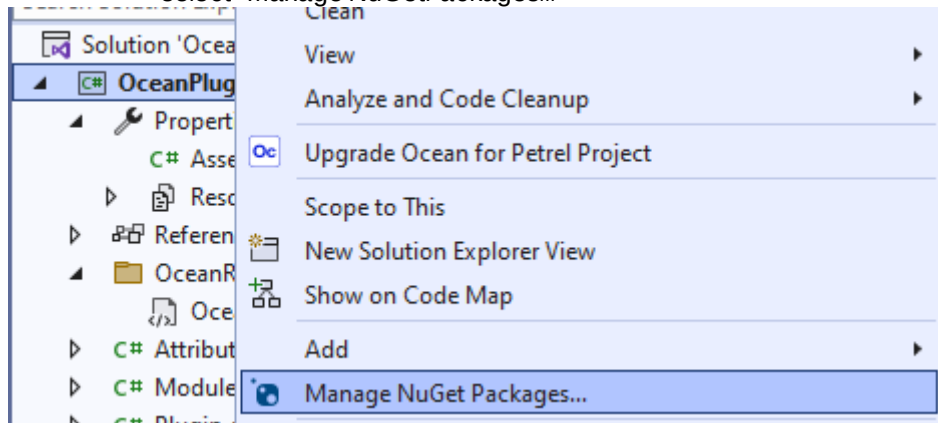


Figure 1: Manage NuGetPackages for Test Project

Then search for "nunit test adapter" and select "NUnitTestAdapter" by Charlie Poole and Terje Sandstrom and click "Install" button for version 2.3.0.

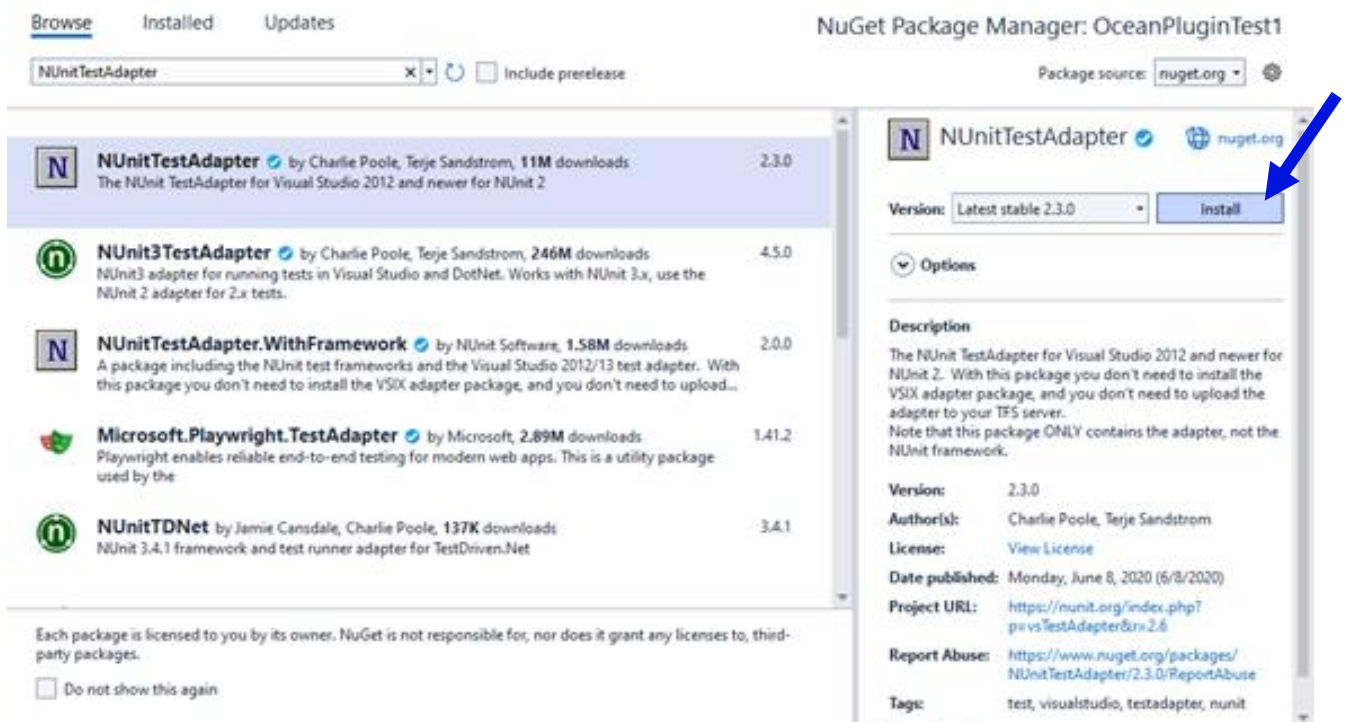


Figure 2: Install NUnitTestAdapter package

13. Known Issue - Unable to create new plug-in using wizard template with Visual Studio 17.6

When using Visual Studio 17.6, if you attempt to create a new Ocean Plug-in, the plug-in may fail to be created and you'll see the following error message:

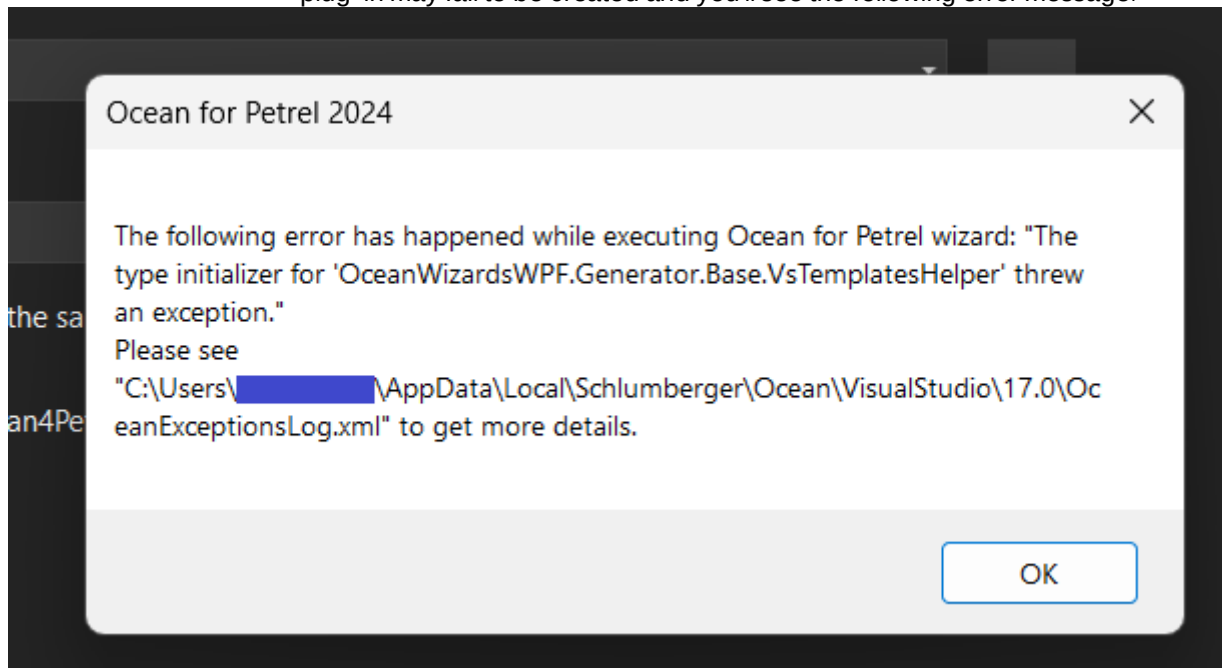


Figure 3: Error message box in Visual Studio 2022 stating 'The type initializer for OceanWizardsWPF.Generator.Base.VBsTemplatesHelper' threw an exception

To resolve this issue, update your Visual Studio to version 17.8 or higher. Then, you should be able to create the plug-in as normal.

14. Assembly versions

Ocean for Petrel assembly versions at the time of the SDK release are as follows:

Assembly	Assembly version
Slb.Ocean.Petrel	21.3.0.0
Slb.Ocean.Petrel.Configuration	21.3.0.0
Slb.Ocean.Petrel.Data.Persistence	21.3.0.0
Slb.Ocean.Petrel.DomainObject.FrameworkModeling	21.3.0.0
Slb.Ocean.Petrel.DomainObjectExtensions	21.3.0.0
Slb.Ocean.Petrel.DomainObjectExtensions.Prestack	21.3.0.0
Slb.Ocean.Petrel.Drilling	21.3.0.0
Slb.Ocean.Petrel.Geology	21.3.0.0
Slb.Ocean.Petrel.Intersect	21.3.0.0
Slb.Ocean.Petrel.Microseismic	21.3.0.0

Slb.Ocean.Petrel.Modeling	21.3.0.0
Slb.Ocean.Petrel.Polygon	21.3.0.0
Slb.Ocean.Petrel.Production	21.3.0.0
Slb.Ocean.Petrel.ReservoirEngineering	21.3.0.0
Slb.Ocean.Petrel.Resources	21.3.0.0
Slb.Ocean.Petrel.Seismic	21.3.0.0
Slb.Ocean.Petrel.Simulation.EclipseKeywords	21.3.0.0
Slb.Ocean.Petrel.StructuralGeology	21.3.0.0
Slb.Ocean.Petrel.Testing	21.3.0.0
Slb.Ocean.Petrel.UI.Controls	21.3.0.0
Slb.Ocean.Petrel.UI.Visualization3D	21.3.0.0
Slb.Ocean.Petrel.UI.WellSection	21.3.0.0
Slb.Ocean.Petrel.UI.Wpf.Controls	21.3.0.0
Slb.Ocean.Petrel.Well	21.3.0.0
Slb.Ocean.Petrel.Well.Treatment	21.3.0.0
Slb.Ocean.Studio.Find.dll	21.1.0.0
Slb.Ocean.Studio.Find.Index.dll	21.1.0.0
Slb.Ocean.Studio.Find.Petrel.dll	21.1.0.0
Slb.Ocean.Studio.Knowledge.Petrel	21.1.0.0

